

In-Class Lab: Spearman's Rank Correlation Test Using Boston Housing Data

Objective:

Students will analyze the relationship between socioeconomic factors (e.g., GDP, population) and suicide rates using the Spearman's Rank Correlation Test. This exercise will reinforce their understanding of monotonic relationships and data cleaning.

Instructions:

1. Download the Dataset:

- Access the dataset from Kaggle: [Boston Housing Prices Dataset](#)
- Load the dataset into a Python environment using `pandas`.

2. Data Exploration and Cleaning:

- **Inspect the data:** Use `head()`, `info()`, and `describe()` to understand the dataset.
- **Handle missing values** (if any): Drop the missing values.

3. Feature Selection for Analysis:

Choose two or three numeric columns to study the relationships.

Suggested pairs:

- **CRIM** (per capita crime rate) vs. **MEDV** (median value of owner-occupied homes).
- **RM** (average number of rooms per dwelling) vs. **LSTAT** (percentage of lower-status population).
- Make sure to choose at least 3 pairs of numeric columns.

4. Perform Spearman's Rank Correlation:

- Use `scipy.stats.spearmanr()` to compute the correlation coefficient and p-value for the selected variables.
- Interpret the results:
 - Is the relationship significant?
 - Is it positive or negative?
 - How strong is the monotonic relationship?

5. Visualization:

- Create scatter plots with trendlines to visually inspect the relationships.
- Use **seaborn** to generate heatmaps of the Spearman correlation matrix for the entire dataset.

6. Submit Your Findings:

- **Report:** Write a brief report summarizing:
 - Steps for cleaning the data.
 - Variables analyzed and why they were chosen.
 - Correlation results and interpretations.
 - Insights and conclusions about the dataset based on the analysis.

Deliverables:

Detailed report with calculations, visualizations, and interpretations.

Grading Criteria:

Correlation Calculation (20%)

Visualizations (Scatterplots, Trendlines) (20%)

Interpretation and Insights (30%)

Report (30%)